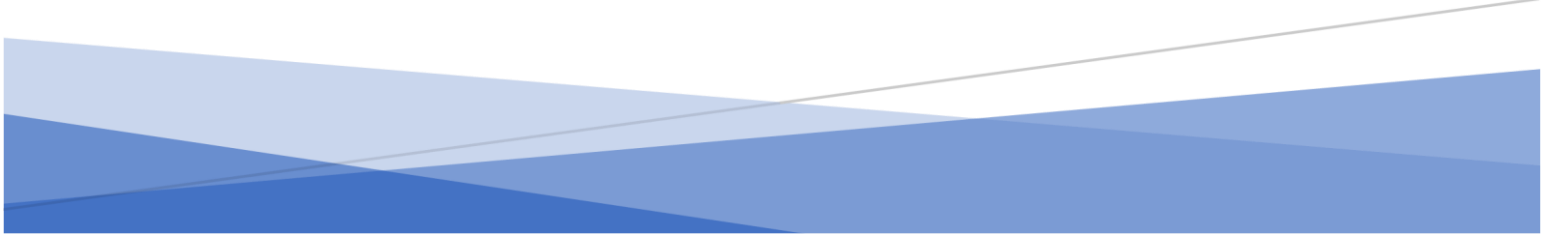


COMP1841 REPORT

Web Development

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GCS240108

Word count: 1529



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GIT link: <https://github.com/Haoiuca/Comp1841-CW.git>

1. Introduction

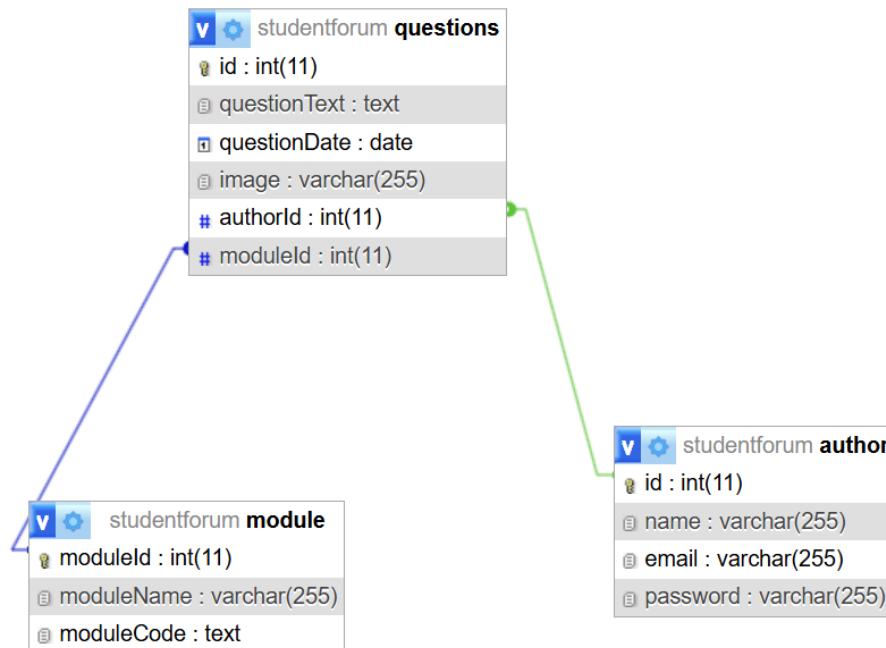
The goal for this project was to create and implement a fully functional “Student Stack Overflow” web application as well as develop an operational “student stack overflow” web platform that uses data (i.e., has functionality). Students can use the application to login securely; post academic related questions; and be able to assign each question to a particular module. Reference images can also be uploaded for support with the assignment of each posted question.

A full implementation of this web-based application was completed using three primary technologies; HTML5, CSS3, PHP, and MySQL. The key consideration when developing the overall structure of the application was the use of the controller/template paradigm. In order to accomplish this, output buffering was utilized (`ob_start()` and `ob_get_clean()`), The separation of the back-end business logic (database query) from the front-end presentation (HTML template) was made possible through the structural decision that is in accordance with the DRY (Don't Repeat Yourself) Principle. As a result of this choice, all parts of the application can utilize one master layout file for presentations; therefore, the system has high scalability as well as being very easy to update and modify.

Throughout the entire development process, Git was used as a method of Version Control. Code was committed frequently, and pushed to a remote GitHub repository so that there would be a safe back-up copy of my files, their organization, and my work.

2. Database Design & Relational Integrity

To ensure data consistency and eliminate redundancy, the database was carefully normalized using the InnoDB storage engine.



The system relies on three interconnected tables:

- **author:** Stores registered user details and hashed passwords.
- **module:** Stores university course codes and names.
- **questions:** The central table storing the user's posts and image paths.

Referential Integrity

To keep data consistent, two Foreign Keys was used in the question table. The FOREIGN KEYS reference both PRIMARY KEYs from the author table and the module table.

The ON DELETE CASCADE rule was applied to both of those FOREIGN KEY references. The reason for this was so that when a user deletes a module from the application, as well as when a user's account is deleted, it will remove ALL related questions to that module or author from the database. This keeps orphaned data out of the database which could result in rendering issues on the front end.

3. Application Security

Security was a primary consideration during development. In particular, they have attempted to mitigate some of the most common web-based attacks to satisfy learning outcome two.

- **Prevention of SQL Injection Attacks:** Never concatenate POST/GET data directly with SQL statements. Raw Post / Get Data can be used with PHP PDO prepared statements. Using the prepare() method, you will bind your variables via an array of

parameters. This ensures the database engine views all user input as simply data, rather than as executable code.

- Hashing Passwords: User passwords are never stored on the server. When you add authors to the database, we take the user's password and salt it, then use PHP's `password_hash()` function to create a one-way hash. On the other hand, when we verify a password, we utilize `password_verify()` located in our `login.php` controller. We ensure user passwords are always secure even if the database is ever breached.
- Protection Of Sessions: The admin/ directory has been fully restricted for unauthorized access to the site. The admin directory utilizes PHP sessions (`session_start()`), and there is a centralized security script called `Check.php` which is included at the top of each admin controller. If the session variable `$_SESSION['loggedin']` is found to be null or empty, the system immediately redirects the unauthorized user to the public login page and completely blocks all access to the "Create", "Update," and "Delete" functions.

4. Legal, Ethical, and Professional Concerns

The process of developing a public-facing website entails compliance with ethical, legal, and professional standards.

- **GDPR/ Data Protection:** Because this application captures personally identifiable data (Name and email address from the authors table) that is subject to privacy regulations such as GDPR; this application will need a clearly stated Privacy policy for the users' benefit, before collecting any data obtain express consent from the user, provide the capability for the user to delete his/her personal data, if they so desire.
- **Accessibility:** Ethically, the web should work for everybody, including people who have a disability, whether it be visual or motor impairment. To assist Screen Readers, the application utilizes Semantic HTML5 Tags to make the site accessible to Screen Reader Software.
- To guarantee that the application operates effectively regardless of the browser type as well as to make sure the design will remain flexible and functional irrespective of the size of the screen display, CSS flexbox and/or grid properties were used, as well as the `<meta name="viewport">` tag within the HTML head of each page.

5. Testing & Implementation

The system was initially tested on a local XAMPP environment to verify CRUD (Create, Read, Update, Delete) functionality.

All Data Manipulation (Create, Update, Delete) operations have been designed to follow architecture standards as stated in the modules. The reason for this is so that these operations are limited to the admin/ directory and therefore are restricted from being accessed by anyone who has not successfully logged into the system. Therefore, the only people who will be able to use the forms required to modify the data base are authenticated users. The public facing pages of the application are strictly read-only for general viewers. Additionally, the Conditional Logic within each page prevents users from editing or deleting anything except questions they created themselves (i.e., those with the same authorId)

Testing the Create functionality and physical image file upload:

Student Stack Overflow

Home

Questions List

Ask a Question

Admin

Login

Contact Us

Ask a New Question

Type your problem description here:

Help me!



Attach a Screenshot (Optional):

Choose File

Screenshot ... 122106.png

Select Academic Module Related to Query:

COMP1841 - Web Programming 1

Publish Question

Student Stack Overflow

Edit

Delete

Question:

Help me!



Testing the Update and Delete functionality in admin panel:

Student Stack Overflow - Admin Panel

[Questions List](#)[Public Site](#)[Manage Modules](#)[Manage Authors](#)[Log Out](#)

Edit Question Entry

Modify your query content:

Help me!
1324



Author:

Alice Smith ▾

Module Assignment:

COMP1648 - Database Systems ▾

Save Changes

Student Stack Overflow - Admin Panel

[Edit](#) [Delete](#)

Question:

Help me! 1324



Module: COMP1648 - Database Systems | Asked by: [Alice Smith](#) | On: 09-08-2026

[Edit](#) [Delete](#)

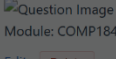
Student Stack Overflow - Admin Panel



Module: COMP1648 - Database Systems | Asked by: Alice Smith | On: 16-10-2026

[Edit](#) [Delete](#)

Question:
Why am I getting a Foreign Key constraint error on my InnoDB table deletion?



Module: COMP1841 - Web Programming 1 | Asked by: Matt Prichard | On: 15-10-2026

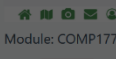
[Edit](#) [Delete](#)

Question:
How do I implement Output Buffering with ob_start() in PHP?

test

Notification

-  You have been notified of a new question.
-  You have been notified of a new question.
-  You have been notified of a new question.
-  You have been notified of a new question.



Module: COMP1773 - User Interface Design | Asked by: Matt Prichard | On: 08-08-2026

[Edit](#) [Delete](#)

Question:
How do I implement Output Buffering with ob_start() in PHP?

weve

Module: COMP1437 - Systems Architecture | Asked by: Alice Smith | On: 31-07-2026

[Edit](#) [Delete](#)

Student Stack Overflow - Admin Panel



Module: COMP1648 - Database Systems | Asked by: Alice Smith | On: 16-10-2026

[Edit](#) [Delete](#)

Question:
Why am I getting a Foreign Key constraint error on my InnoDB table deletion?



Module: COMP1841 - Web Programming 1 | Asked by: Matt Prichard | On: 15-10-2026

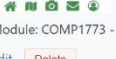
[Edit](#) [Delete](#)

Question:
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test

Notification

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-  You have been notified of a new question.
-  You have been notified of a new question.
-  You have been notified of a new question.



Module: COMP1773 - User Interface Design | Asked by: Matt Prichard | On: 08-08-2026

[Edit](#) [Delete](#)

Question:
How do I implement Output Buffering with ob_start() in PHP?

Testing the Contact Form:

Student Stack Overflow

[Home](#) [Questions List](#) [Ask a Question](#) [Admin](#) [Login](#) [Contact Us](#)

Contact Us

Have a question about the Student Stack Overflow? Send us a message!

Your Email Address:

Your Message:



6. Conclusion and Future Recommendation

The design and construction of the Student Stack Overflow prototype have been successful in meeting the needs outlined in the COMP1841 course specifications. Specifically, the primary goal of creating a working, data-driven web application has been achieved by the careful integration of HTML5, CSS3, PHP 8, and MySQL.

One of the major successes of this project has been the very strict compliance with the Controller/Template architecture. Through the separation of back-end PHP business logic and front-end HTML presentation layers, the code base is both modular and easy to maintain. Additionally, by utilizing output buffering (`ob_start()`) potential issues arising due to HTTP Header Conflicts were avoided, allowing dynamic content to be injected into a centralized Master Layout and therefore satisfying DRY (Don't Repeat Yourself) principles.

From a data perspective, the MySQL database used to support the prototype was designed to utilize the InnoDB storage engine. Therefore, it provides robust relational integrity through foreign key relationships between the authors, modules, and questions tables. To ensure security throughout development, the sole usage of PHP Data Objects (PDO) with prepared statements effectively eliminated the possibility of SQL Injection attacks. Furthermore, try-catch blocks were utilized to handle database error scenarios without exposing potentially sensitive system information to users.

Additionally, the application demonstrates secure access to an administrative section. By separating data manipulation (create, read, update delete (CRUD)) actions from behind a session-based login boundary, the system prevents unauthorized changes to data while at

the same time allows users who are logged in to perform read-only operations on data within their own sessions.

Future Recommendation:

Although the current version of the prototype fulfills the initial requirements, numerous upgrades need to occur before making it available to a larger university population:

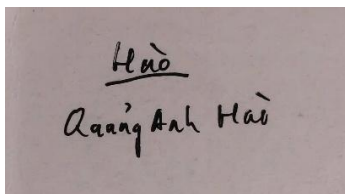
- **Secure Password Hashing Using Cryptography:** One of the first significant updates should include the implementation of PHP's native `password_hash()`, utilizing the `bcrypt` hashing algorithm, along with `password_verify()` for securely storing passwords in addition to preventing malicious entities from accessing your database.
- **SQL Pagination and Keyword Search Functionality:** As additional student questions are added to the repository, a single `select *` query will become resource-intensive on servers when trying to fetch all records. Adding SQL limit and offset functionality for paginating results will significantly enhance scalability. Similarly, adding a keyword search feature will greatly enhance usability and scalability.
- **Safe Rich Text Input Editing:** Currently, all text entered by students is sanitized so that cross-site scripting (XSS) does not occur. However, integrating a safe WYSIWYG editor such as TinyMCE and/or a server side HTML Purifier will provide students with a means to create formatted code blocks or highlight specific text areas without negatively affecting application security.

Declaration of AI Use

Please append this page to the end of your assignment when you have used AI during the process of undertaking the assignment to acknowledge how you have used it.

I have used AI while undertaking my assignment in the following ways:

- To develop research questions on the topic – YES
- To create an outline of the topic – YES
- To explain concepts – YES
- To support my use of language – YES
- To summarise the following articles/resources:



Haò
Quang Anh Hải